

Samaneh Mohammadi

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WORK EXPERIENCE

Associate Researcher (Part-time)

Sep 2026 – Present

[School of Informatics, The University of Edinburgh](#) · *Edinburgh, United Kingdom*

- **Trustworthiness in LLM-based systems for high-stakes domains:** LLM-based systems are increasingly deployed in **high-stakes domains** such as hiring pipelines. This project studies how **trustworthy** these systems are in such settings with a focus on their **fairness** and **security**. It is funded by and carried out in collaboration with Amazon (ongoing).

PhD Researcher (Full-time)

Mar 2026 – Present

[School of Informatics, The University of Edinburgh](#) · *Edinburgh, United Kingdom*

- **Bias and fairness in large language models (LLMs):** Bias in LLMs is **concentrated in particular components**. This work first locates which attention heads and feed-forward layers carry it, then fine-tunes only those components against a fairness-aware training objective (ongoing).
- **Privacy–fairness tensions in distributed ML:** differential privacy often widens performance gaps across demographic groups, creating fairness issues. To address this, we designed a **group-aware model architecture** that reduces performance gaps across groups under differential privacy, without sharing group information. Publication under review.

PhD Researcher & Developer (Full-time)

Sep 2021 – Feb 2026

[RISE Research Institutes of Sweden](#) · *Västerås, Sweden*

Conducted research and development across two EU-funded projects:

[DAIS \(Distributed Artificial Intelligent Systems\)](#) projects (EU flagship project, 47 partners):

- **Private machine unlearning framework:** letting participants privately remove their data's influence from a collaboratively trained model, removing the reliance on a central server. Publication ([CIKM '25](#)).
- **Scalable and efficient privacy preservation in distributed ML:** where a model is trained collaboratively across many devices and only model parameters are shared for aggregation, we proposed **efficient homomorphic and functional encryption frameworks** that encrypt each device's model updates locally before they are shared, while remaining practical at scale. Publications ([PMIJ '25](#), [Workshop at NeurIPS '24](#), [ISPEC '23](#)).
- **Real-world model deployment:** Built a physical testbed for collaboratively training speech emotion recognition models across heterogeneous edge devices, measuring efficiency, fairness, and privacy trade-offs under real deployment conditions. Publication ([IJCNN '25](#)).

[DADAP \(Digitizing and Automating the Diagnostic Psychiatric Assessment Process\)](#) projects:

- **Healthcare AI on sensitive data:** benchmarked a range of ML models, including recent transformer architectures, on clinical mental-health tabular records collected in Swedish hospitals, and provided a broad evaluation to provide reliable decision support for clinicians. Publication ([PharML at ECML PKDD '26](#)).

Technology Engineer / Software Developer (Full-time)

Sep 2020 – Aug 2021

[Openinside Co. W.L.L](#) · *Manama, Bahrain (Remote)*

- **Enterprise software development (ERP):** built new Odoo modules end to end (Python backend, JavaScript front end), supporting finance and supply-chain operations for business clients.
- **Systems integration:** connected those modules to clients existing finance and supply-chain systems, keeping data and processes consistent across both.

Research Assistant (MSc Thesis)

Sep 2017 – Oct 2020

[CoinLab, University of Tehran](#) · *Tehran, Iran*

- **Anomaly detection in information networks:** developed graph-based methods that flag suspicious nodes by tracking their activity and how it changes over time, an approach applicable to fraud and abuse detection in social networks.

SELECTED PUBLICATIONS (200+ CITATIONS)

- Mohammadi, S.*, Tsouvalas, V.* et al., [EFU: Enforcing Federated Unlearning via Functional Encryption](#), the 34th ACM International Conference on Information and Knowledge Management (CIKM '25).
- Tsouvalas, V.*, Mohammadi, S.* et al., [EncCluster: Scalable Functional Encryption through Weight Clustering and Probabilistic Filters](#), Pervasive and Mobile Computing Journal (PMIJ '25).

- **Mohammadi, S.** et al., [Empirical Analysis of Asynchronous Federated Learning on Heterogeneous Devices: Efficiency, Fairness, and Privacy Trade-offs](#), **International Joint Conference on Neural Networks (IJCNN '25)**.
- Tsouvalas, V.*, **Mohammadi, S.*** et al., [Bringing Functional Encryption to Federated Foundation Models](#), **Workshop at NeurIPS '24**.
- **Mohammadi, S.** et al., [Balancing Privacy and Performance in Federated Learning: A Systematic Literature Review on Methods and Metrics](#), **Journal of Parallel and Distributed Computing (JPDC '24)**.
- **Mohammadi, S.** et al., [Balancing Privacy and Accuracy in Federated Learning for Speech Emotion Recognition](#), **18th conference on computer science and intelligence systems (FedCSIS '23)**, **Professor Zdzisław Pawlak Award**.

* Equal contribution.

EDUCATION

Ph.D. in Cyber Security, Privacy & Trust

Sep 2021 – Expected Dec 2026

[School of Informatics, University of Edinburgh](#) · *Edinburgh, United Kingdom*

Thesis: Reconciling Privacy with Fairness and Accountability in Federated Learning under Practical Constraints. Supervisor: [Dr. Marc Juarez](#).

Note: My PhD started at RISE Research Institutes of Sweden, enrolled at Mälardalen University; transferred to the University of Edinburgh, where I will complete my degree.

M.Sc. in Information Technology Engineering

Sep 2017 – Oct 2020

[University of Tehran](#) · *Tehran, Iran*

Thesis: Anomaly Detection in Dynamic Information Networks. Supervisor: [Dr. Mostafa Salehi](#). GPA: 17.70/20, **ranked 2nd in class**.

B.Sc. in Information Technology Engineering

Sep 2012 – Sep 2016

[Azarbaijan Shahid Madani University](#) · *Tabriz, Iran*

GPA: 17.04/20, **ranked 1st in class**.

HONOURS & AWARDS

- **PhD Scholarship**, School of Informatics, University of Edinburgh, 2026.
- **ACM SIG Travel Grant** to present at CIKM, Seoul, 2025.
- **Professor Zdzisław Pawlak Award**, FedCSIS 2023.
- **Ranked 68th among 19,598 candidates** (top 0.4%) in the national M.Sc. entrance exam in Information Technology Engineering, Iran, 2017.

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, C++, C#, MATLAB.
- **Frameworks & Libraries:** PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, Flower.
- **Machine Learning:** Federated, distributed, and on-device learning; LLM fine-tuning (LoRA / PEFT); transformer models for language, vision, and tabular data (e.g. LLaMA, BERT, ViT).
- **Privacy & Security:** Differential privacy, homomorphic encryption, functional encryption, machine unlearning.
- **Model Evaluation:** Benchmarking ML models and LLMs for trustworthiness, including privacy leakage, security risk, and fairness, across language, vision, and tabular data.
- **Tools, Infrastructure & Databases:** Git, Docker, Linux, cloud platforms; GPU cluster computing, Slurm job scheduling (srun), PostgreSQL, MS SQL Server.
- **Governance & Compliance:** GDPR, EU AI Act.

TALKS & ACADEMIC SERVICE

- **Program Committee** at [IEEE Conference on Federated Learning Technologies and Applications \(FLTA\)](#), 2025 and 2026.
- **Program Committee** at [CFAgentic @ ICML 2025 Workshop](#).
- **Invited Speaker** at [Flower Monthly](#) for “Scalable Functional Encryption in Federated Learning,” hosted by Prof. Nicholas Lane (University of Cambridge / co-founder of Flower Labs), Sep 2024.
- **Invited Speaker** at [MegaData Summer School on Federated Machine Learning](#) for “Balancing Privacy and Performance in Federated Learning,” organiser: University of Tartu, Aug 2024.
- **Conference Presentations:** ECML PKDD '26 (Naples), CIKM '25 (Seoul), IJCNN '25 (Rome), NeurIPS '24 (Vancouver), ISPEC '23 (Copenhagen), FedCSIS '23 (Warsaw).